

ISO71xxCC 4242-V_{PK} Small-Footprint Low-Power Triple and Quad Channels Digital Isolators

1 Features

- Maximum Signaling Rate: 50 Mbps (With 5-V Supplies)
- Robust Design With Integrated Noise Filter
- Default Output Low Option (Suffix F)
- Low Power Consumption, Typical I_{CC} per Channel (With 3.3-V Supplies):
 - ISO7131: 1.5 mA at 1 Mbps, 2.6 mA at 25 Mbps
 - ISO7140: 1 mA at 1 Mbps, 2.3 mA at 25 Mbps
 - ISO7141: 1.3 mA at 1 Mbps, 2.6 mA at 25 Mbps
- Low Propagation Delay: 23-ns Typical (3.3-V Supplies)
- Wide Temperature Range: –40°C to 125°C
- 50-kV/μs Transient Immunity, Typical
- Long Life With SiO₂ Isolation Barrier
- Operates from 2.7-V, 3.3-V, and 5-V Supply and Logic Levels
- Small QSOP-16 Package
- Safety and Regulatory Approvals
 - 2500-V_{RMS} Isolation for 1 minute per UL 1577
 - 4242-V_{PK} Isolation per DIN V VDE V 0884-10 (VDE V 0884-10):2006-12, 566 V_{PK} Working Voltage
 - CSA Component Acceptance Notice 5A, IEC 60950-1 and IEC 61010-1 End Equipment Standards
 - CQC Certification per GB 4943.1-2011

2 Applications

- General-Purpose Isolation
 - Industrial Fieldbus
 - Profibus
 - Modbus™
 - DeviceNet Data Buses
 - RS-232, RS-485
 - Serial Peripheral Interface

3 Description

ISO7131, ISO7140, and ISO7141 devices provide galvanic isolation up to 2500 V_{RMS} for 1 minute per UL and 4242 V_{PK} per VDE. ISO7131 has three channels with two forward and one reverse-direction channels. ISO7140 and ISO7141 are quad-channel isolators; ISO7140 has four forward channels, ISO7141 has three forward and one reverse-direction channels. These devices are capable of 50-Mbps maximum data rate with 5-V supplies and 40-Mbps maximum data rate with 3.3-V or 2.7-V supplies, with integrated filters on the inputs for noise-prone applications. The suffix F indicates that default output state is low; otherwise, the default output state is high (see Table 3).

Each isolation channel has a logic input and output buffer separated by a silicon dioxide (SiO₂) insulation barrier. Used with isolated power supplies, these devices prevent noise currents on a data bus or other circuits from entering the local ground and interfering with or damaging sensitive circuitry. The devices have TTL input thresholds and can operate from 2.7-V, 3.3-V, and 5-V supplies. All inputs are 5-V tolerant when supplied from a 2.7-V or 3.3-V supply.

Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
ISO7131CC	SSOP (16)	4.90 mm × 3.90 mm
ISO7140CC		
ISO7140FCC		
ISO7141CC		
ISO7141FCC		

(1) For all available packages, see the orderable addendum at the end of the datasheet.

Simplified Schematic

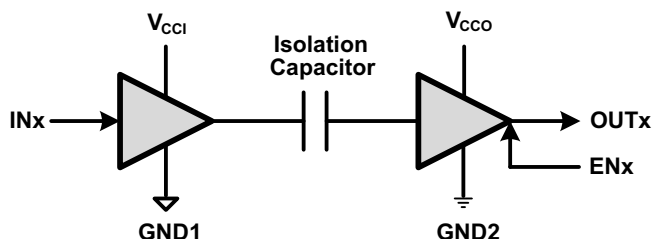


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4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision E (September 2013) to Revision F	Page
• Added <i>Pin Configuration and Functions</i> section, <i>ESD Ratings</i> table, <i>Feature Description</i> section, <i>Device Functional Modes</i> , <i>Application and Implementation</i> section, <i>Power Supply Recommendations</i> section, <i>Layout</i> section, <i>Device and Documentation Support</i> section, and <i>Mechanical, Packaging, and Orderable Information</i> section	1
• VDE Standard changed to DIN V VDE V 0884-10 (VDE V 0884-10):2006-12	1

Changes from Revision D (August 2013) to Revision E	Page
• Changed From: 2500 V_{RMS} Isolation for 1 minute per UL 1577 (Approval Pending) To: (Approved)	1
• Added note1 to the AVAILABLE OPTIONS table	17
• Changed Figure 15	18
• Changed From: Basic Insulation To: Basic Insulation, Altitude ≤ 5000m, Tropical Climate, 250 VRMS maximum working voltage in the Regulatory Information table	19
• Changed File number: E181974 (approval pending) To: File number: E181974 in the Regulatory Information table	19
• Changed the title of Figure 21 , Figure 22 , and Figure 23 to include "PRBS 2 ¹⁶ - 1"	23

Changes from Revision C (July 2013) to Revision D	Page
• Added Safety List item "GB 4943.1-2011 and GB 8898:2011 CQC Certification (Approval Pending)"	1
• Added Figure 2	12
• Deleted "Product Preview" From the AVAILABLE OPTIONS table	17
• Changed the REGULATORY INFORMATION, added column for CQC	19

Changes from Revision B (June 2013) to Revision C
Page

• Changed Feature From: ISO7140: TBD at 1 Mbps, TBD at 25 Mbps To: ISO7140: 1 mA at 1 Mbps, 2.3 mA at 25 Mbps..	1
• Added text to the Description: "All inputs are 5V tolerant when supplied from a 2.7V or 3.3V supply."	1
• Deleted the Product Status table.....	1
• Changed the SAFETY and REGULATORY APPROVALS	1
• Changed the ABSOLUTE MAXIMUM RATINGS table	5
• Changed the SWITCHING CHARACTERISTICS table, Input glitch rejection time.	7
• Changed the SWITCHING CHARACTERISTICS table, Input glitch rejection time.	8
• Changed the SWITCHING CHARACTERISTICS table, Input glitch rejection time.	8
• Changed ISO7140 in the SUPPLY CURRENT table From: TBD To: values.....	9
• Changed ISO7140 in the SUPPLY CURRENT table From: TBD To: values.....	10
• Changed ISO7140 in the SUPPLY CURRENT table From: TBD To: values.....	11
• Changed Figure 1 X-axis scale	12
• Changed the AVAILABLE OPTIONS table.....	17

Changes from Revision A (June 2013) to Revision B
Page

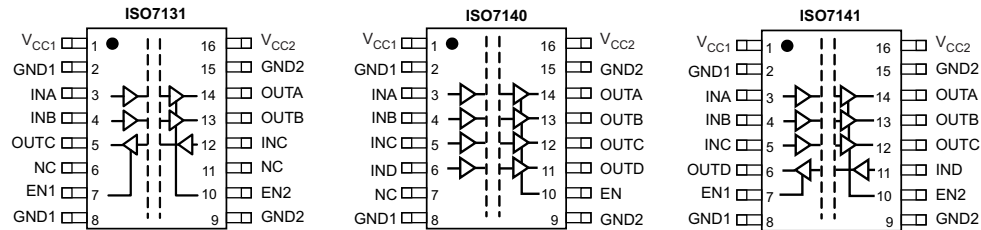
• Changed device ISO7141CC From: Product Preview To: Released in the Product Status table	1
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Changes from Original (April 2013) to Revision A
Page

• Changed the Simplified Schematic, added ground symbols.....	1
• Changed the SWITCHING CHARACTERISTICS table, Input glitch rejection time. Values by device.	7
• Changed the SWITCHING CHARACTERISTICS table, Input glitch rejection time. Values by device.	8
• Changed the SWITCHING CHARACTERISTICS table, Input glitch rejection time. Values by device.	8
• Added Figure 3	12

5 Pin Configuration and Functions

**16-Pin
SSOP Package
Top View**



Pin Functions

PIN				I/O	DESCRIPTION
NAME	ISO7131	ISO7140	ISO7141		
EN	—	10	—	I	Output enable. All output pins are enabled when EN is high or disconnected and disabled when EN is low.
EN1	7	—	7	I	Output enable 1. Output pins on side-1 are enabled when EN1 is high or disconnected and disabled when EN1 is low.
EN2	10	—	10	I	Output enable 2. Output pins on side-2 are enabled when EN2 is high or disconnected and disabled when EN2 is low.
GND1	2,8	2,8	2,8	—	Ground connection for V_{CC1}
GND2	9,15	9,15	9,15	—	Ground connection for V_{CC2}
INA	3	3	3	I	Input, channel A
INB	4	4	4	I	Input, channel B
INC	12	5	5	I	Input, channel C
IND	—	6	11	I	Input, channel D
NC	6,11	7	—	—	No Connect pins are floating with no internal connection
OUTA	14	14	14	O	Output, channel A
OUTB	13	13	13	O	Output, channel B
OUTC	5	12	12	O	Output, channel C
OUTD	—	11	6	O	Output, channel D
V_{CC1}	1	1	1	—	Power supply, V_{CC1}
V_{CC2}	16	16	16	—	Power supply, V_{CC2}

6 Specifications

6.1 Absolute Maximum Ratings⁽¹⁾

		MIN	MAX	UNIT
V _{CC1} , V _{CC2}	Supply voltage ⁽²⁾	–0.5	6	V
IN _x , EN _x , OUT _x	Voltage	–0.5	V _{CC} + 0.5 ⁽³⁾	V
I _O	Output current	–15	15	mA
T _J	Maximum junction temperature		150	°C
T _{stg}	Storage temperature	–65	150	°C

- (1) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under [Recommended Operating Conditions](#) is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) All voltage values except differential I/O bus voltages are with respect to the local ground terminal (GND1 or GND2) and are peak voltage values.
- (3) Maximum voltage must not exceed 6 V

6.2 ESD Ratings

		VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±4000
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±1500

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

		MIN	NOM	MAX	UNIT
V _{CC1} , V _{CC2}	Supply voltage	2.7		5.5	V
I _{OH}	High-level output current (V _{CC} ≥ 3.0 V)	–4			mA
	High-level output current (V _{CC} < 3.0 V)	–2			
I _{OL}	Low-level output current			4	mA
V _{IH}	High-level input voltage	2		5.5	V
V _{IL}	Low-level input voltage	0		0.8	
t _{ui}	Input pulse duration (V _{CC} ≥ 4.5V)	20			ns
t _{ui}	Input pulse duration (V _{CC} < 4.5V)	25			
1 / t _{ui}	Signaling rate (V _{CC} ≥ 4.5V)	0		50	Mbps
1 / t _{ui}	Signaling rate (V _{CC} < 4.5V)	0		40	
T _A	Ambient temperature	–40	25	125	°C
T _J	Junction temperature	–40		136	

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾		ISO7131, ISO714x	UNIT
		DBQ	
		16 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	104.5	°C/W
R _{θJC(top)}	Junction-to-case(top) thermal resistance	57.8	°C/W
R _{θJB}	Junction-to-board thermal resistance	46.8	°C/W
ψ _{JT}	Junction-to-top characterization parameter	18.3	°C/W
ψ _{JB}	Junction-to-board characterization parameter	46.4	°C/W

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report, [SPRA953](#).

6.5 Power Dissipation Ratings

	TEST CONDITIONS	VALUE	UNIT
P _D	Device power dissipation V _{CC1} = V _{CC2} = 5.5 V, T _J = 150°C, C _L = 15 pF Input a 25-MHz, 50% duty cycle square wave	150	mW

6.6 Electrical Characteristics: V_{CC1} and V_{CC2} at 5 V ±10%

V_{CC1} and V_{CC2} at 5 V ± 10% (over recommended operating conditions unless otherwise noted.)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
V _{OH}	I _{OH} = –4 mA; see Figure 10	V _{CCO} ⁽¹⁾ – 0.5	4.8		V
	I _{OH} = –20 μA; see Figure 10	V _{CCO} ⁽¹⁾ – 0.1	5		
V _{OL}	I _{OL} = 4 mA; see Figure 10		0.2	0.4	V
	I _{OL} = 20 μA; see Figure 10		0	0.1	
V _{I(HYS)}	Input threshold voltage hysteresis		450		mV
I _{IH}	High-level input current	V _{IH} = V _{CC} at INx or ENx		10	μA
I _{IL}	Low-level input current	V _{IL} = 0 V at INx or ENx	–10		μA
CMTI	Common-mode transient immunity	V _I = V _{CC} or 0 V; see Figure 13	25	75	kV/μs

(1) V_{CCO} is the supply voltage, V_{CC1} or V_{CC2}, for the output channel that is being measured.

6.7 Electrical Characteristics: V_{CC1} and V_{CC2} at 3.3 V ±10%

V_{CC1} and V_{CC2} at 3.3 V ±10% (over recommended operating conditions unless otherwise noted.)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
V _{OH}	I _{OH} = –4 mA; see Figure 10	V _{CCO} ⁽¹⁾ – 0.5	3		V
	I _{OH} = –20 μA; see Figure 10	V _{CCO} ⁽¹⁾ – 0.1	3.3		
V _{OL}	I _{OL} = 4 mA; see Figure 10		0.2	0.4	V
	I _{OL} = 20 μA; see Figure 10		0	0.1	
V _{I(HYS)}	Input threshold voltage hysteresis		425		mV
I _{IH}	High-level input current	V _{IH} = V _{CC} at INx or ENx		10	μA
I _{IL}	Low-level input current	V _{IL} = 0 V at INx or ENx	–10		μA
CMTI	Common-mode transient immunity	V _I = V _{CC} or 0 V; see Figure 13	25	50	kV/μs

(1) V_{CCO} is the supply voltage, V_{CC1} or V_{CC2}, for the output channel that is being measured.

6.8 Electrical Characteristics: V_{CC1} and V_{CC2} at 2.7 V

V_{CC1} and V_{CC2} at 2.7 V (over recommended operating conditions unless otherwise noted.)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{OH} High-level output voltage	$I_{OH} = -2 \text{ mA}$; see Figure 10	$V_{CCO}^{(1)} - 0.3$	2.5		V
	$I_{OH} = -20 \text{ }\mu\text{A}$; see Figure 10	$V_{CCO}^{(1)} - 0.1$	2.7		
V_{OL} Low-level output voltage	$I_{OL} = 4 \text{ mA}$; see Figure 10		0.2	0.4	V
	$I_{OL} = 20 \text{ }\mu\text{A}$; see Figure 10		0	0.1	
$V_{I(HYS)}$ Input threshold voltage hysteresis			350		mV
I_{IH} High-level input current	$V_{IH} = V_{CC}$ at INx or ENx			10	μA
I_{IL} Low-level input current	$V_{IL} = 0 \text{ V}$ at INx or ENx	-10			μA
CMTI Common-mode transient immunity	$V_I = V_{CC}$ or 0 V; see Figure 13	25	50		kV/ μs

(1) V_{CCO} is the supply voltage, V_{CC1} or V_{CC2} , for the output channel that is being measured.

6.9 Switching Characteristics: V_{CC1} and V_{CC2} at 5 V $\pm 10\%$

V_{CC1} and V_{CC2} at 5 V $\pm 10\%$ (over recommended operating conditions unless otherwise noted.)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{PLH} , t_{PHL} Propagation delay time	See Figure 10	12	19	35	ns
$PWD^{(1)}$ Pulse width distortion $ t_{PHL} - t_{PLH} $				3	
$t_{sk(o)}^{(2)}$ Channel-to-channel output skew time	Same-direction channels			2	ns
	Opposite-direction channels			4	
$t_{sk(pp)}^{(3)}$ Part-to-part skew time				12	ns
t_r Output signal rise time	See Figure 10		2		ns
t_f Output signal fall time			2		ns
t_{PHZ} , t_{PLZ} Disable propagation delay, high/low-to-high impedance output	See Figure 11		6	10	ns
t_{PZH} , t_{PZL} Enable propagation delay, high impedance-to-high/low output			5	10	ns
t_{fs} Fail-safe output delay time from input data or power loss	See Figure 12		9.5		μs
t_{GR} Input glitch rejection time			11		ns

(1) Also known as pulse skew

(2) $t_{sk(o)}$ is the skew between outputs of a single device with all driving inputs connected together and the outputs switching in the same direction while driving identical loads.

(3) $t_{sk(pp)}$ is the magnitude of the difference in propagation delay times between any terminals of different devices switching in the same direction while operating at identical supply voltages, temperature, input signals, and loads.

6.10 Switching Characteristics: V_{CC1} and V_{CC2} at 3.3 V $\pm 10\%$

V_{CC1} and V_{CC2} at 3.3 V $\pm 10\%$ (over recommended operating conditions unless otherwise noted.)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{PLH} , t_{PHL}	Propagation delay time	See Figure 10	15	23	45	ns
PWD ⁽¹⁾	Pulse width distortion $ t_{PHL} - t_{PLH} $				3	
$t_{sk(o)}$ ⁽²⁾	Channel-to-channel output skew time	Same-direction Channels			2	ns
		Opposite-direction Channels			4	
$t_{sk(pp)}$ ⁽³⁾	Part-to-part skew time				19	ns
t_r	Output signal rise time	See Figure 10		2.5		ns
t_f	Output signal fall time			2.5		ns
t_{PHZ} , t_{PLZ}	Disable propagation delay, from high/low to high-impedance output	See Figure 11		6.5	15	ns
t_{PZH} , t_{PZL}	Enable propagation delay, from high-impedance to high/low output			6.5	15	ns
t_{fs}	Fail-safe output delay time from input data or power loss	See Figure 12		8		μ s
t_{GR}	Input glitch rejection time			12.5		ns

(1) Also known as pulse skew

(2) $t_{sk(o)}$ is the skew between outputs of a single device with all driving inputs connected together and the outputs switching in the same direction while driving identical loads.

(3) $t_{sk(pp)}$ is the magnitude of the difference in propagation delay times between any terminals of different devices switching in the same direction while operating at identical supply voltages, temperature, input signals and loads.

6.11 Switching Characteristics: V_{CC1} and V_{CC2} at 2.7 V

V_{CC1} and V_{CC2} at 2.7 V (over recommended operating conditions unless otherwise noted.)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{PLH} , t_{PHL}	Propagation delay time	See Figure 10	15	27	50	ns
PWD ⁽¹⁾	Pulse width distortion $ t_{PHL} - t_{PLH} $				3	
$t_{sk(o)}$ ⁽²⁾	Channel-to-channel output skew time	Same-direction Channels			2	ns
		Opposite-direction Channels			4	
$t_{sk(pp)}$ ⁽³⁾	Part-to-part skew time				22	ns
t_r	Output signal rise time	See Figure 10		3		ns
t_f	Output signal fall time			3		ns
t_{PHZ} , t_{PLZ}	Disable propagation delay, from high/low to high-impedance output	See Figure 11		9	15	ns
t_{PZH} , t_{PZL}	Enable propagation delay, from high-impedance to high/low output			9	15	ns
t_{fs}	Fail-safe output delay time from input data or power loss	See Figure 12		8.5		μ s
t_{GR}	Input glitch rejection time			14		ns

(1) Also known as pulse skew

(2) $t_{sk(o)}$ is the skew between outputs of a single device with all driving inputs connected together and the outputs switching in the same direction while driving identical loads.

(3) $t_{sk(pp)}$ is the magnitude of the difference in propagation delay times between any terminals of different devices switching in the same direction while operating at identical supply voltages, temperature, input signals, and loads.

6.12 Supply Current: V_{CC1} and V_{CC2} at 5 V $\pm 10\%$

V_{CC1} and V_{CC2} at 5 V $\pm 10\%$ (over recommended operating conditions unless otherwise noted.)

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
ISO7131						
I _{CC1}	Disable	EN1 = EN2 = 0 V	2.2	3.7	mA	
I _{CC2}			3.7	5		
I _{CC1}	DC to 1 Mbps	DC signal: V _I = V _{CC} or 0 V AC signal: All channels switching with square-wave clock input; C _L = 15 pF	2.2	3.7	mA	
I _{CC2}			3.7	5		
I _{CC1}	10 Mbps		3.4	4.8		
I _{CC2}			4.9	6.6		
I _{CC1}	25 Mbps		4.9	6.6		
I _{CC2}			6.8	9		
I _{CC1}	50 Mbps		7.1	10		
I _{CC2}			10.5	13		
ISO7140						
I _{CC1}	Disable	EN = 0 V	0.6	1.2	mA	
I _{CC2}			4.6	7		
I _{CC1}	DC to 1 Mbps	DC Signal: V _I = V _{CC} or 0 V, AC Signal: All channels switching with square wave clock input; C _L = 15 pF	0.6	1.3	mA	
I _{CC2}			4.8	7		
I _{CC1}	10 Mbps		1.4	2.2		
I _{CC2}			6.9	9.2		
I _{CC1}	25 Mbps		2.7	3.9		
I _{CC2}			10.3	13.5		
I _{CC1}	50 Mbps		4.7	6.5		
I _{CC2}			15.6	21		
ISO7141						
I _{CC1}	Disable	EN1 = EN2 = 0V	2.5	4.2	mA	
I _{CC2}			4.2	7		
I _{CC1}	DC to 1 Mbps	DC signal: V _I = V _{CC} or 0 V, AC signal: All channels switching with square wave clock input; C _L = 15 pF	2.5	4.2	mA	
I _{CC2}			4.2	7		
I _{CC1}	10 Mbps		3.8	5.3		
I _{CC2}			6.2	9.6		
I _{CC1}	25 Mbps		5.6	7.5		
I _{CC2}			9.2	13		
I _{CC1}	50 Mbps		8.4	11.2		
I _{CC2}			14	18.5		

6.13 Supply Current: V_{CC1} and V_{CC2} at 3.3 V $\pm 10\%$

V_{CC1} and V_{CC2} at 3.3 V $\pm 10\%$ (over recommended operating conditions unless otherwise noted.)

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
ISO7131						
I _{CC1}	Disable	EN1 = EN2 = 0 V	1.9	2.7	mA	
I _{CC2}			2.6	3.8		
I _{CC1}	DC to 1 Mbps	DC signal: V _I = V _{CC} or 0 V AC signal: All channels switching with square-wave clock input; C _L = 15 pF	1.9	2.7	mA	
I _{CC2}			2.6	3.8		
I _{CC1}	10 Mbps		2.4	3.5		
I _{CC2}			3.5	4.7		
I _{CC1}	25 Mbps		3.2	4.6		
I _{CC2}			4.7	6.2		
I _{CC1}	40 Mbps		5	7		
I _{CC2}			7	9		
ISO7140						
I _{CC1}	Disable	EN = 0 V	0.3	0.7	mA	
I _{CC2}			3.6	5.2		
I _{CC1}	DC to 1 Mbps	DC signal: V _I = V _{CC} or 0 V, AC signal: All channels switching with square-wave clock input; C _L = 15 pF	0.4	0.8	mA	
I _{CC2}			3.7	5.3		
I _{CC1}	10 Mbps		0.9	1.4		
I _{CC2}			5.1	6.8		
I _{CC1}	25 Mbps		1.7	2.4		
I _{CC2}			7.3	10		
I _{CC1}	40 Mbps		2.4	3.7		
I _{CC2}			9.4	13		
ISO7141						
I _{CC1}	Disable	EN1 = EN2 = 0 V	2	3.1	mA	
I _{CC2}			3.2	4.9		
I _{CC1}	DC to 1 Mbps	DC signal: V _I = V _{CC} or 0 V, AC signal: All channels switching with square-wave clock input; C _L = 15 pF	2	3.1	mA	
I _{CC2}			3.2	4.9		
I _{CC1}	10 Mbps		2.8	3.8		
I _{CC2}			4.5	6.1		
I _{CC1}	25 Mbps		4	5.2		
I _{CC2}			6.4	8.3		
I _{CC1}	40 Mbps		5	8		
I _{CC2}			8.2	11.6		

6.14 Supply Current: V_{CC1} and V_{CC2} at 2.7 V

V_{CC1} and V_{CC2} at 2.7 V (over recommended operating conditions unless otherwise noted.)

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
ISO7131						
I _{CC1}	Disable	EN1 = EN2 = 0 V	1.2		2.4	mA
I _{CC2}			2.3		3.3	
I _{CC1}	DC to 1 Mbps	DC signal: V _I = V _{CC} or 0 V AC signal: All channels switching with square-wave clock input; C _L = 15 pF	1.2		2.4	mA
I _{CC2}			2.3		3.3	
I _{CC1}	10 Mbps		2.1		3	
I _{CC2}			2.9		4	
I _{CC1}	25 Mbps		3		3.8	
I _{CC2}			4		5.2	
I _{CC1}	40 Mbps		4.2		5.3	
I _{CC2}			5.8		7	
ISO7140						
I _{CC1}	Disable	EN = 0 V	0.2		0.4	mA
I _{CC2}			3.2		4.7	
I _{CC1}	DC to 1 Mbps	DC signal: V _I = V _{CC} or 0 V, AC signal: All channels switching with square-wave clock input; C _L = 15 pF	0.2		0.5	mA
I _{CC2}			3.4		4.8	
I _{CC1}	10 Mbps		0.6		1	
I _{CC2}			4.5		6.3	
I _{CC1}	25 Mbps		1.2		1.8	
I _{CC2}			6.2		8	
I _{CC1}	40 Mbps		1.8		2.6	
I _{CC2}			8		11	
ISO7141						
I _{CC1}	Disable	EN1 = EN2 = 0 V	1.6		2.6	mA
I _{CC2}			2.8		4.1	
I _{CC1}	DC to 1 Mbps	DC signal: V _I = V _{CC} or 0 V, AC signal: All channels switching with square-wave clock input; C _L = 15 pF	1.6		2.6	mA
I _{CC2}			2.8		4.1	
I _{CC1}	10 Mbps		2.3		3.2	
I _{CC2}			3.8		5	
I _{CC1}	25 Mbps		3.3		4.2	
I _{CC2}			5.4		6.8	
I _{CC1}	40 Mbps		4.3		5.8	
I _{CC2}			6.9		9.2	

6.15 Typical Characteristics

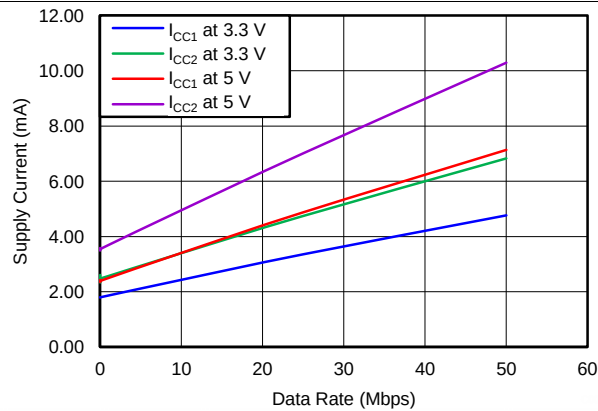


Figure 1. ISO7131 Supply Current for All Channels vs Data Rate

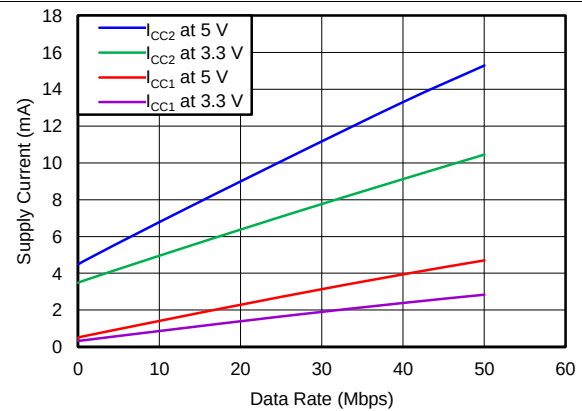


Figure 2. ISO7140 Supply Current for All Channels vs Data Rate

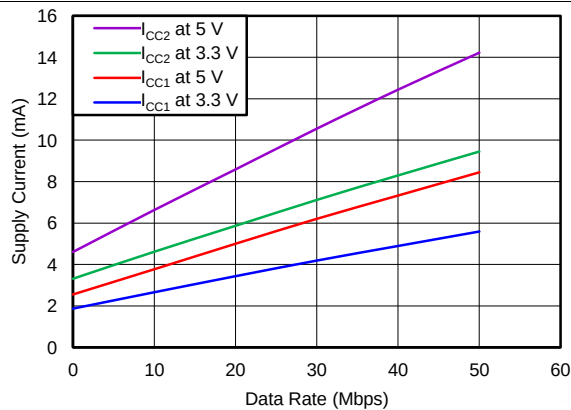


Figure 3. ISO7141 Supply Current for All Channels vs Data Rate

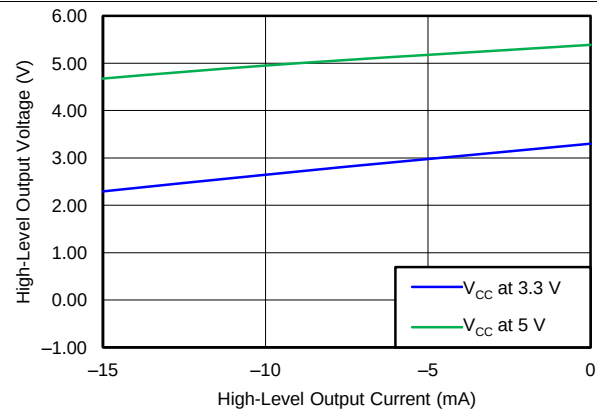


Figure 4. High-Level Output Voltage vs High-Level Output Current

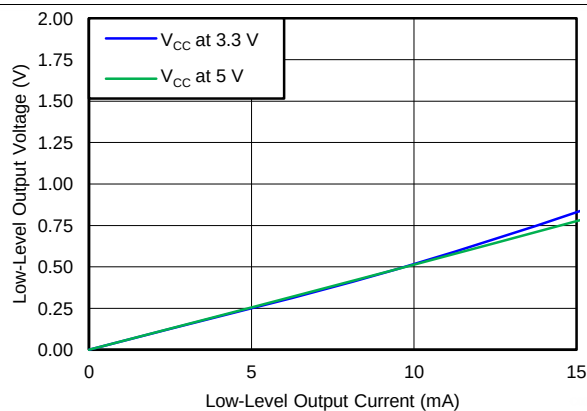


Figure 5. Low-Level Output Voltage vs Low-Level Output Current

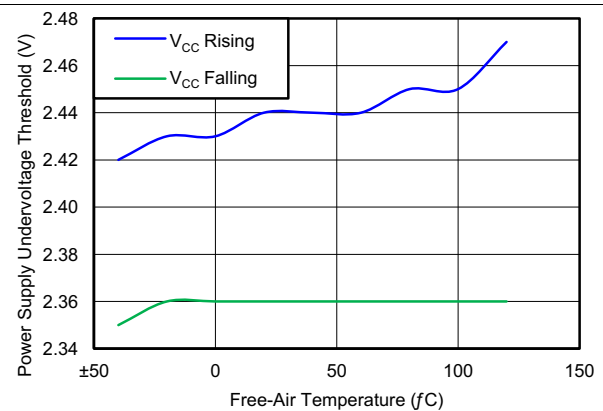


Figure 6. VCC Undervoltage Threshold vs Free-Air Temperature

Typical Characteristics (continued)

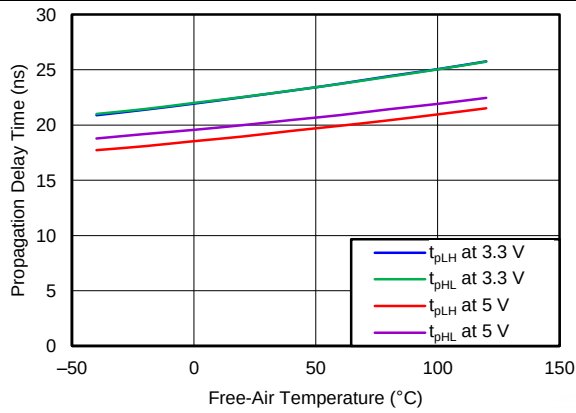


Figure 7. Propagation Delay Time vs Free-Air Temperature

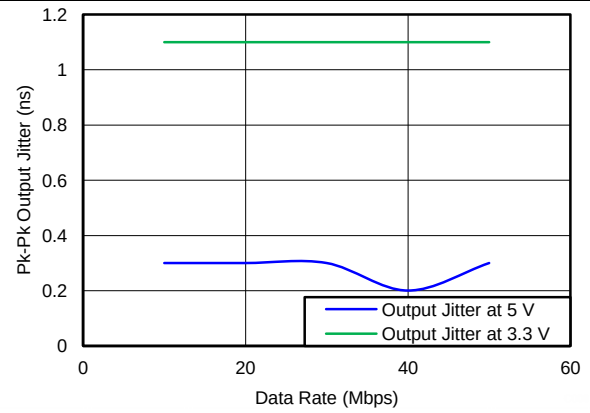


Figure 8. Output Jitter vs Data Rate

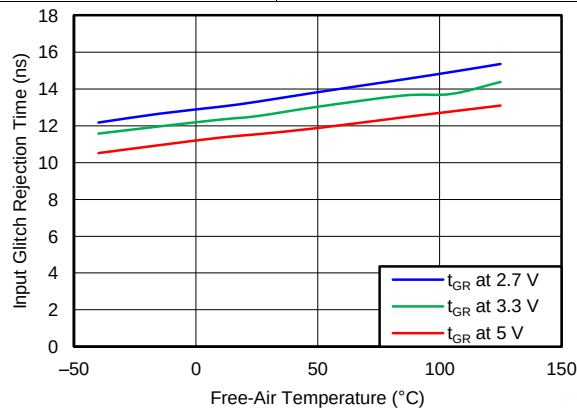
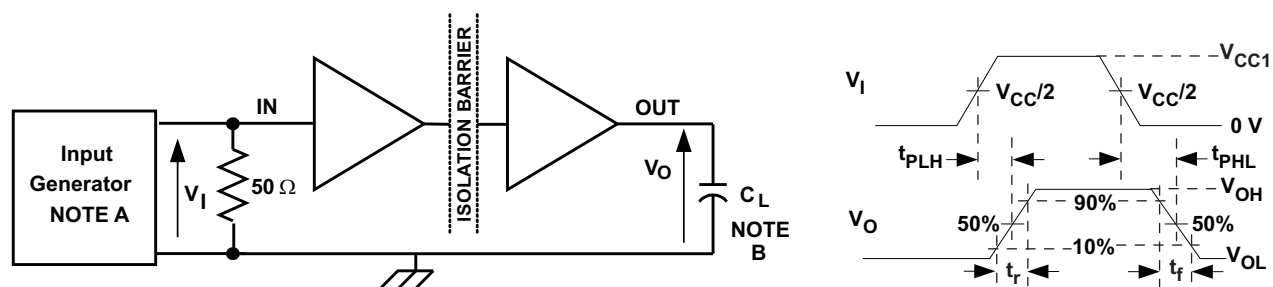


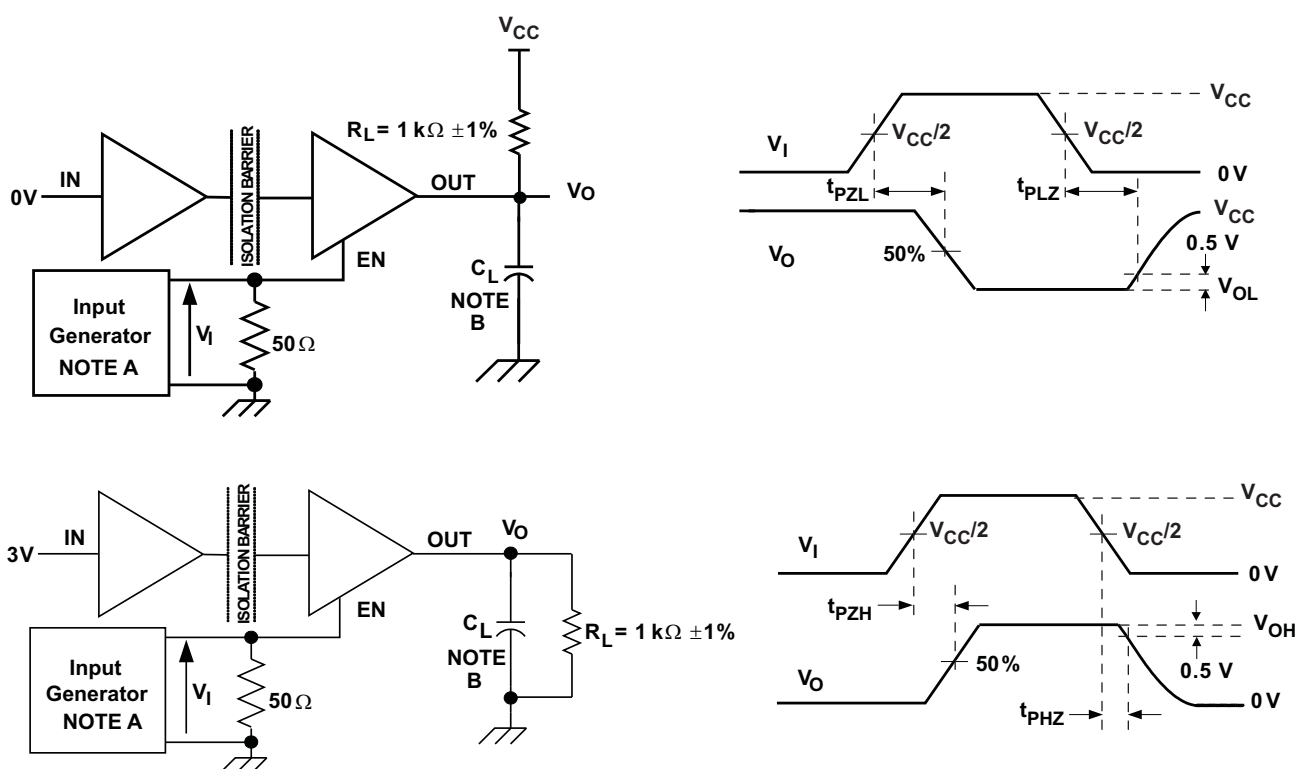
Figure 9. Input Glitch Rejection vs Free-Air Temperature

7 Parameter Measurement Information



- A. The input pulse is supplied by a generator having the following characteristics: PRR $\leq$ 50 kHz, 50% duty cycle, $t_r \leq 3$ ns, $t_f \leq 3$ ns, $Z_O = 50 \Omega$. At the input, a 50- Ω resistor is required to terminate the input-generator signal. It is not needed in an actual application.
- B. $C_L = 15$ pF and includes instrumentation and fixture capacitance within $\pm 20\%$.

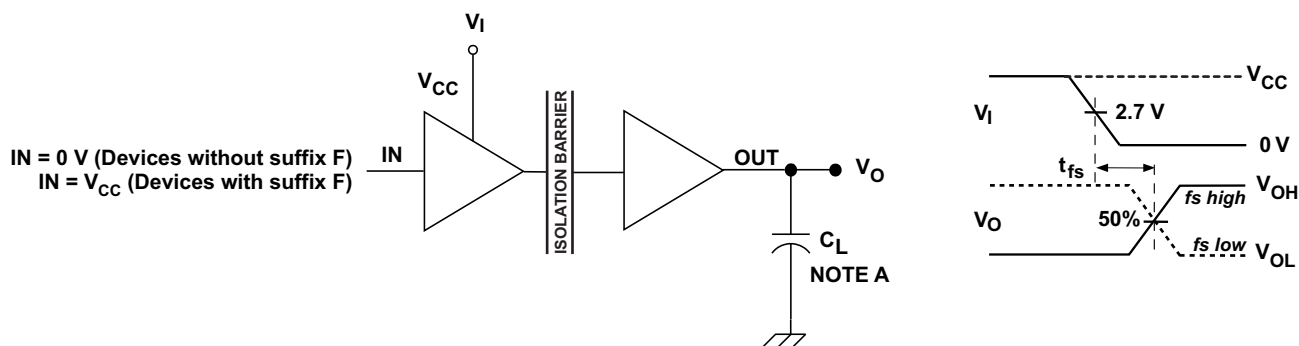
Figure 10. Switching-Characteristics Test Circuit and Voltage Waveforms



- A. The input pulse is supplied by a generator having the following characteristics: PRR $\leq$ 50 kHz, 50% duty cycle, $t_r \leq 3$ ns, $t_f \leq 3$ ns, $Z_O = 50 \Omega$.
- B. $C_L = 15$ pF and includes instrumentation and fixture capacitance within $\pm 20\%$.

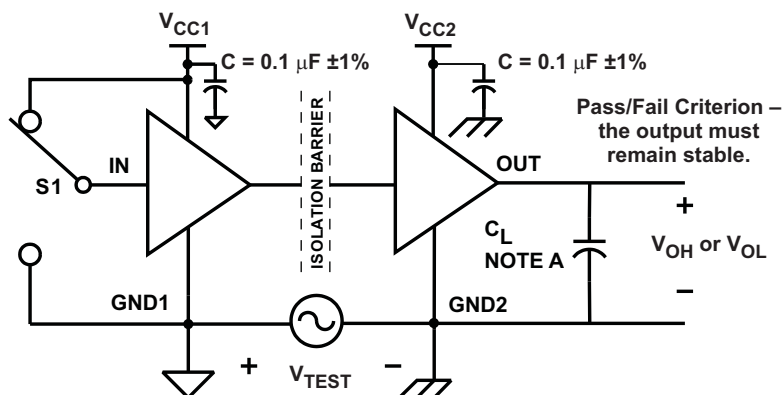
Figure 11. Enable/Disable Propagation Delay-Time Test Circuit and Waveform

Parameter Measurement Information (continued)



A. $C_L = 15$ pF and includes instrumentation and fixture capacitance within $\pm 20\%$.

Figure 12. Failsafe Delay-Time Test Circuit and Voltage Waveforms



A. $C_L = 15$ pF and includes instrumentation and fixture capacitance within $\pm 20\%$.

Figure 13. Common-Mode Transient Immunity Test Circuit

8 Detailed Description

8.1 Overview

The isolator in [Figure 14](#) is based on a capacitive isolation barrier technique. The I/O channel of the device consists of two internal data channels, a high-frequency channel (HF) with a bandwidth from 100 kbps up to 150 Mbps, and a low-frequency channel (LF) covering the range from 100 kbps down to DC. In principle, a single-ended input signal entering the HF-channel is split into a differential signal through the inverter gate at the input. The following capacitor-resistor networks differentiate the signal into transients, which then are converted into differential pulses by two comparators. The comparator outputs drive a NOR-gate flip-flop whose output feeds an output multiplexer. A decision logic (DCL) at the driving output of the flip-flop measures the durations between signal transients. If the duration between two consecutive transients exceeds a certain time limit, (as in the case of a low-frequency signal), the DCL forces the output-multiplexer to switch from the high- to the low-frequency channel.

Because low-frequency input signals require the internal capacitors to assume prohibitively large values, these signals are pulse-width modulated (PWM) with the carrier frequency of an internal oscillator, thus creating a sufficiently high frequency signal, capable of passing the capacitive barrier. As the input is modulated, a low-pass filter (LPF) is needed to remove the high-frequency carrier from the actual data before passing it on to the output multiplexer.

8.2 Functional Block Diagram

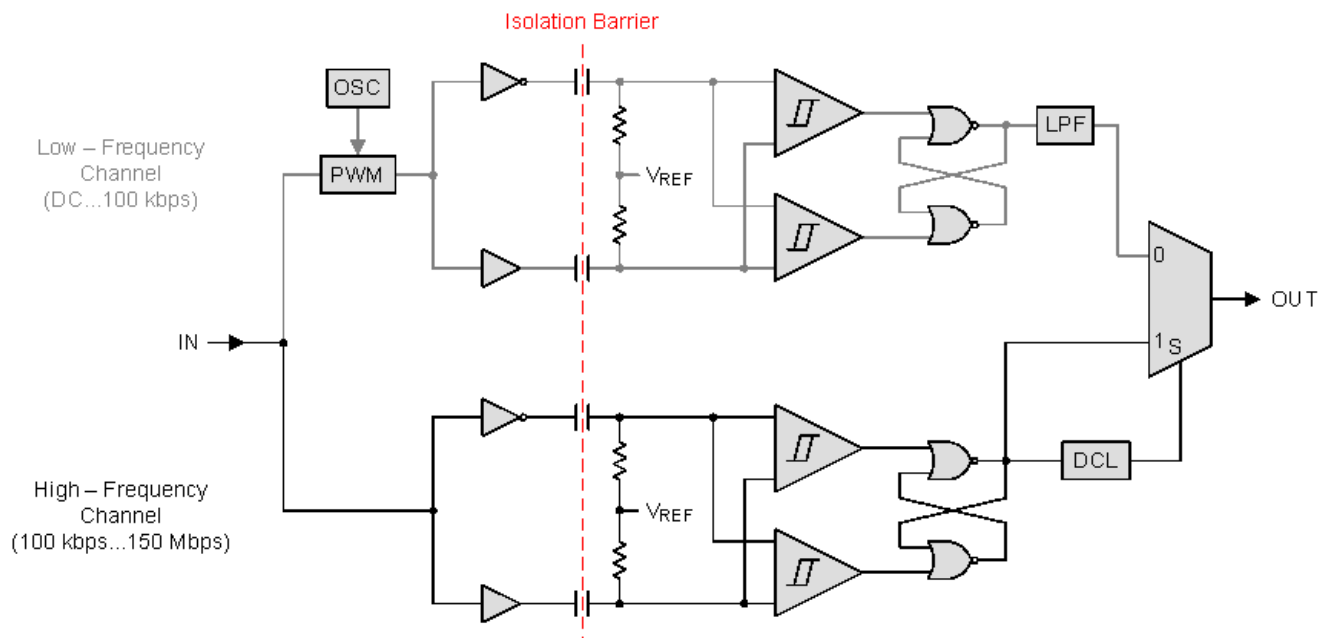


Figure 14. Conceptual Block Diagram of a Digital Capacitive Isolator

8.3 Feature Description

Table 1. Product Features

PRODUCT	RATED ISOLATION	INPUT THRESHOLD	DEFAULT OUTPUT	MAX DATA RATE and INPUT FILTER	CHANNEL DIRECTION	
ISO7131CC	4242 V _{PK} ⁽¹⁾	1.5-V TTL (CMOS compatible)	High	50 Mbps, with noise filter integrated	2 forward, 1 reverse	
ISO7140CC			Low		4 forward, 0 reverse	
ISO7140FCC						
ISO7141CC						High
ISO7141FCC						Low

(1) See [Regulatory Information](#) for detailed Isolation Ratings.

8.3.1 Insulation and Safety-Related Specifications

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V _{IOTM}	Maximum transient overvoltage per DIN V VDE V 0884-10 (VDE V 0884-10):2006-12				4242	V _{PK}
V _{IORM}	Maximum working voltage per DIN V VDE V 0884-10 (VDE V 0884-10):2006-12				566	V _{PK}
V _{ISO}	Isolation Voltage per UL 1577	V _{TEST} = V _{ISO} , t = 60 sec (qualification)			2500	V _{RMS}
		V _{TEST} = 1.2 * V _{ISO} , t = 1 sec (100% production)			3000	
V _{PR}	Input-to-output test voltage per DIN V VDE V 0884-10 (VDE V 0884-10):2006-12	After Input/Output safety test subgroup 2/3, V _{PR} = V _{IORM} × 1.2, t = 10 s, Partial discharge < 5 pC			679	V _{PK}
		Method a, After environmental tests subgroup 1, V _{PR} = V _{IORM} × 1.6, t = 10 s, Partial discharge < 5 pC			906	
		Method b1, 100% production test, V _{PR} = V _{IORM} × 1.875, t = 1 s, Partial discharge < 5 pC			1061	
L(I01)	Minimum air gap (clearance)	Shortest terminal to terminal distance through air	3.7			mm
L(I02)	Minimum external tracking (creepage)	Shortest terminal to terminal distance across the package surface	3.7			mm
	Minimum internal gap (internal clearance)	Distance through the insulation	0.014			mm
	Pollution degree			2		
CTI	Tracking resistance (comparative tracking index)	DIN IEC 60112 / VDE 0303 Part 1	≥400			V
R _{IO} ⁽¹⁾	Isolation Resistance, Input to Output	V _{IO} = 500 V, T _A = 25°C		>10 ¹²		Ω
		V _{IO} = 500 V, 100°C ≤ T _A ≤ T _A max		>10 ¹¹		
C _{IO} ⁽¹⁾	Barrier capacitance, input to output	V _I = 0.4 sin (2πft), f = 1 MHz		2.3		pF
C _I ⁽²⁾	Input capacitance	V _I = V _{CC} /2 + 0.4 sin (2πft), f = 1 MHz, V _{CC} = 5 V		2.8		pF

(1) All pins on each side of the barrier tied together creating a two-terminal device.

(2) Measured from input pin to ground.

NOTE

Creepage and clearance requirements should be applied according to the specific equipment isolation standards of an application. Care should be taken to maintain the creepage and clearance distance of a board design to ensure that the mounting pads of the isolator on the printed-circuit-board (PCB) do not reduce this distance.

Creepage and clearance on a PCB become equal in certain cases. Techniques such as inserting grooves and/or ribs on a PCB are used to help increase these specifications.

Table 2. IEC 60664-1 Ratings Table

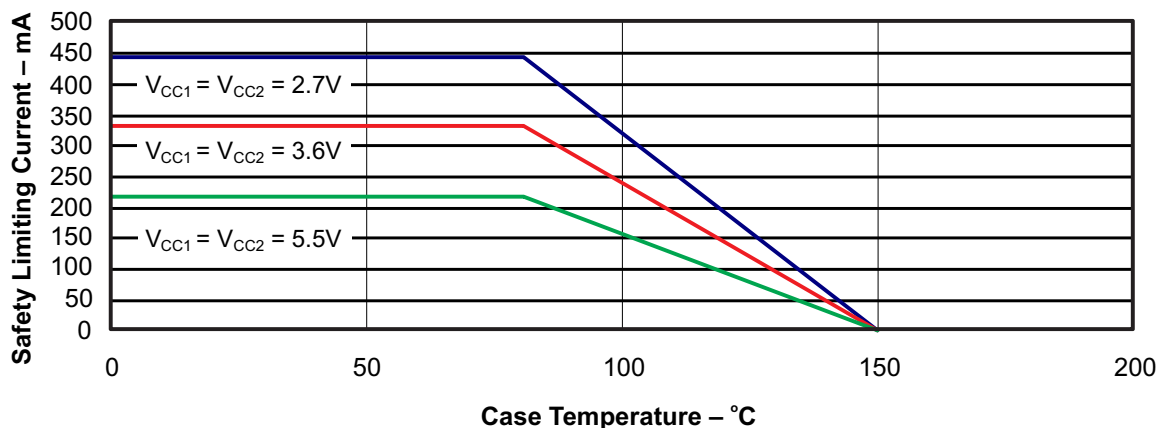
PARAMETER	TEST CONDITIONS	SPECIFICATION
Basic Isolation Group	Material Group	II
Installation classification	Rated mains voltage $\leq 150\text{ V}_{\text{RMS}}$	I–IV
	Rated mains voltage $\leq 300\text{ V}_{\text{RMS}}$	I–III
	Rated mains voltage $\leq 400\text{ V}_{\text{RMS}}$	I–II

8.3.1.1 Safety Limiting Values

Safety limiting intends to prevent potential damage to the isolation barrier upon failure of input or output circuitry. A failure of the IO can allow low resistance to ground or the supply and, without current limiting, dissipate sufficient power to overheat the die and damage the isolation barrier, potentially leading to secondary system failures.

PARAMETER	TEST CONDITIONS			MIN	TYP	MAX	UNIT
I_S Safety input, output, or supply current	DBQ-16	$R_{\theta JA} = 104.5^\circ\text{C/W}$, $V_I = 5.5\text{V}$, $T_J = 150^\circ\text{C}$, $T_A = 25^\circ\text{C}$				217	mA
		$R_{\theta JA} = 104.5^\circ\text{C/W}$, $V_I = 3.6\text{V}$, $T_J = 150^\circ\text{C}$, $T_A = 25^\circ\text{C}$				332	
		$R_{\theta JA} = 104.5^\circ\text{C/W}$, $V_I = 2.7\text{V}$, $T_J = 150^\circ\text{C}$, $T_A = 25^\circ\text{C}$				443	
T_S Maximum case temperature						150	$^\circ\text{C}$

The safety-limiting constraint is the absolute-maximum junction temperature specified in the [Absolute Maximum Ratings](#)⁽¹⁾ table. The power dissipation and junction-to-air thermal impedance of the device installed in the application hardware determines the junction temperature. The assumed junction-to-air thermal resistance in the [Thermal Information](#) table is that of a device installed on a high-K test board for leaded surface-mount packages. The power is the recommended maximum input voltage times the current. The junction temperature is then the ambient temperature plus the power times the junction-to-air thermal resistance.


Figure 15. DBQ-16 θ_{JC} Thermal Derating Curve

(1) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under [Recommended Operating Conditions](#) is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

8.3.1.2 Regulatory Information

VDE	UL	CSA	CQC
Certified according to DIN V VDE V 0884-10 (VDE V 0884-10):2006-12 and DIN EN 61010-1	Recognized under UL 1577 Component Recognition Program	Approved under CSA Component Acceptance Notice 5A, IEC 60950-1, and IEC 61010-1	Certified according to GB 4943.1-2011
Basic Insulation Maximum Transient Overvoltage, 4242 V _{PK} Maximum Working Voltage, 566 V _{PK}	Single protection, 2500 V _{RMS} ⁽¹⁾	Reinforced Insulation per CSA 60950-1-03 and IEC 60950-1 (2nd Ed.), 185 V _{RMS} maximum working voltage Basic Insulation per CSA 60950-1-03 and IEC 60950-1 (2nd Ed.), 370 V _{RMS} maximum working voltage Reinforced Insulation per CSA 61010-1-12 and IEC 61010-1 (3rd Edition), 150 V _{RMS} maximum working voltage Basic Insulation per CSA 61010-1-12 and IEC 61010-1 (3rd Edition), 300 V _{RMS} maximum working voltage	Basic Insulation, Altitude ≤ 5000m, Tropical Climate, 250 V _{RMS} maximum working voltage
Certificate number: 40016131	File number: E181974	Master contract number: 220991	Certificate number: CQC14001109540

(1) Production tested ≥ 3000 Vrms for 1 second in accordance with UL 1577.

8.4 Device Functional Modes

Table 3. Function Table⁽¹⁾

V _{CCI}	V _{CCO}	INPUT (IN _x)	OUTPUT ENABLE (EN _x)	OUTPUT (OUT _x)	
				ISO71xxCC	ISO71xxFCC
PU	PU	H	H or open	H	H
		L	H or open	L	L
		X	L	Z	Z
		Open	H or open	H	L
PD	PU	X	H or open	H	L
PD	PU	X	L	Z	Z
PU	PD	X	X	Undetermined	Undetermined

(1) V_{CCI} = Input-side V_{CC}; V_{CCO} = Output-side V_{CC}; PU = Powered Up (V_{CC} ≥ 2.7 V); PD = Powered Down (V_{CC} ≤ 2.1 V); X = Irrelevant; H = High Level; L = Low Level; Z = High Impedance

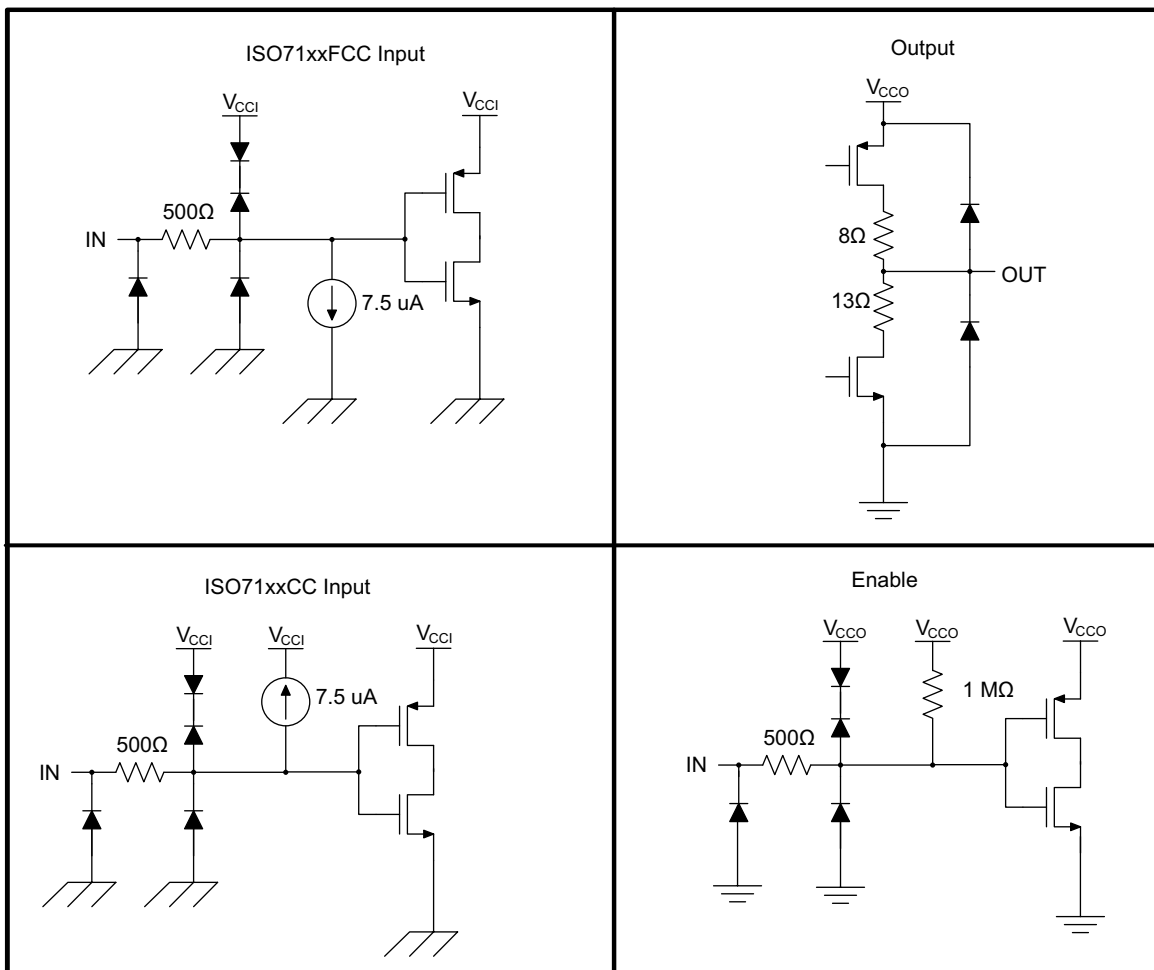


Figure 16. Device I/O Schematics

9 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

ISO71xx use single-ended TTL-logic switching technology. Its supply voltage range is from 3 V to 5.5 V for both supplies, V_{CC1} and V_{CC2} . When designing with digital isolators, it is important to note that due to the single-ended design structure, digital isolators do not conform to any specific interface standard and are only intended for isolating single-ended CMOS or TTL digital signal lines. The isolator is typically placed between the data controller (that is, μC or UART), and a data converter or a line transceiver, regardless of the interface type or standard.

9.2 Typical Applications

9.2.1 Isolated Data Acquisition System for Process Control

ISO71xx combined with TI's precision analog-to-digital converter and mixed signal micro-controller can create an advanced isolated data acquisition system as shown in Figure 17.

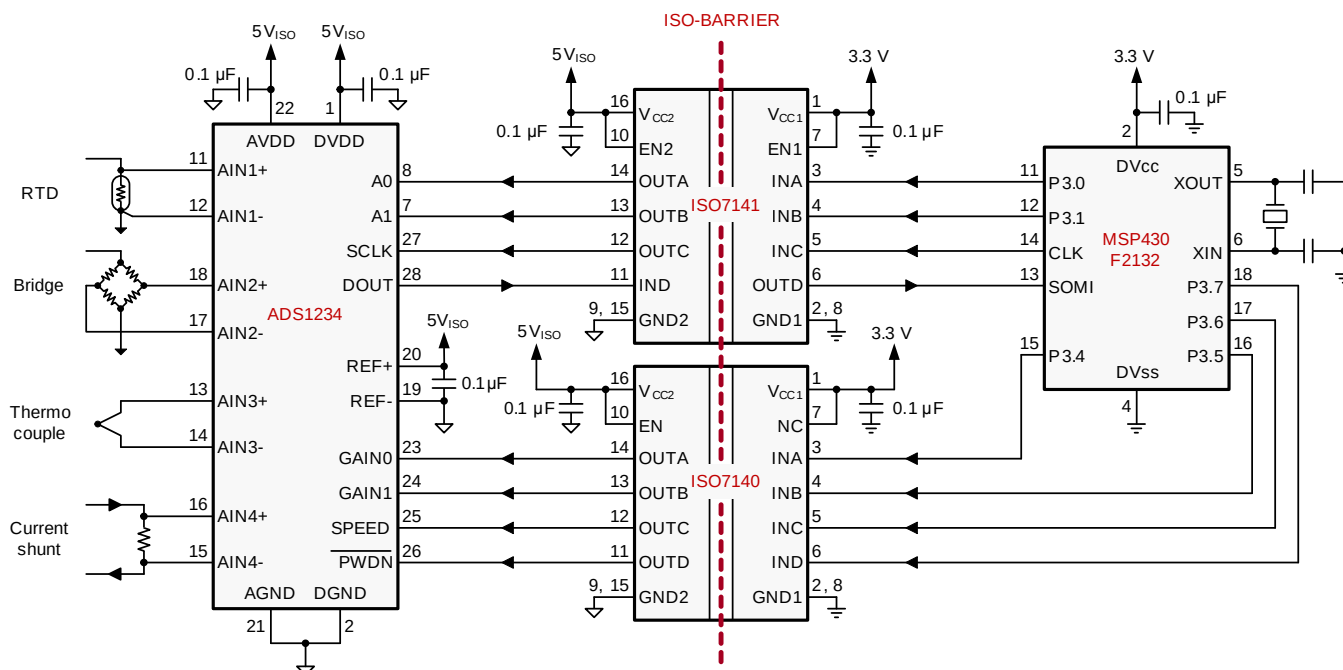


Figure 17. Isolated Data Acquisition System for Process Control

Typical Applications (continued)

9.2.1.1 Design Requirements

Unlike optocouplers, which need external components to improve performance, provide bias, or limit current, ISO71xx only needs two external bypass capacitors to operate.

9.2.1.2 Detailed Design Procedure

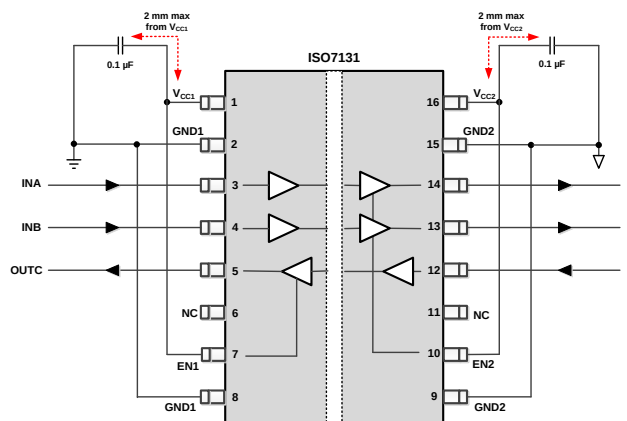


Figure 18. Typical ISO7131 Circuit Hook-up

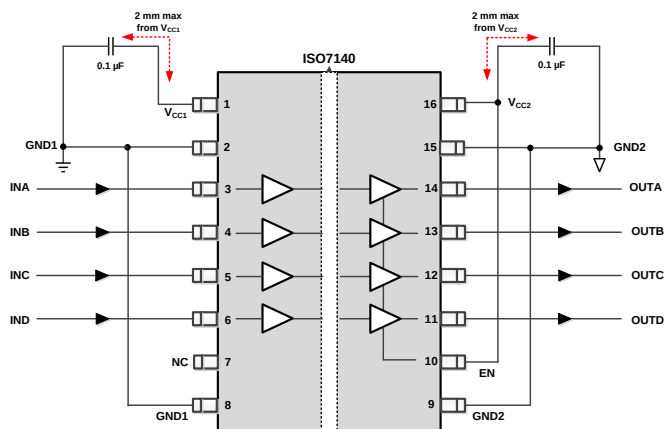


Figure 19. Typical ISO7140 Circuit Hook-up

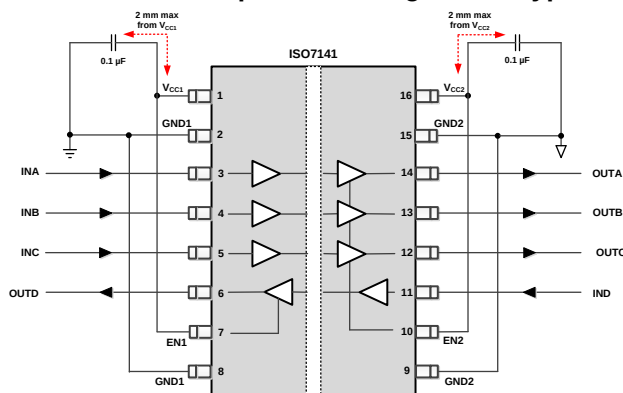


Figure 20. Typical ISO7141 Circuit Hook-up

Typical Applications (continued)

9.2.1.3 Application Curves

Typical eye diagrams of ISO71xx (see [Figure 21](#), [Figure 22](#), and [Figure 23](#)) indicate low jitter and wide open eye at the maximum data rate.

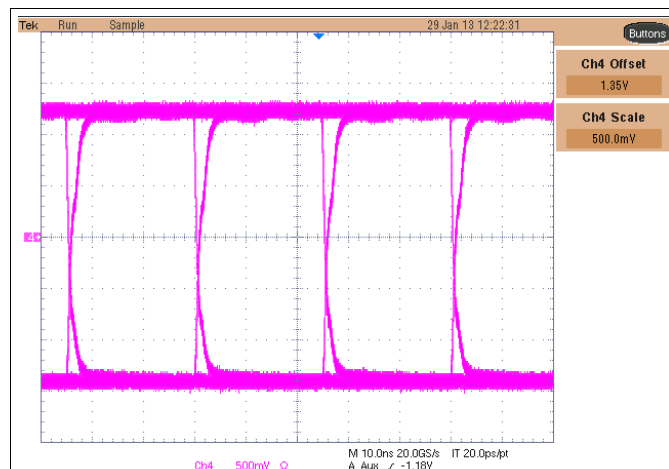


Figure 21. Typical Eye Diagram at 40 MBPS, PRBS 2¹⁶ - 1, 2.7-V Operation

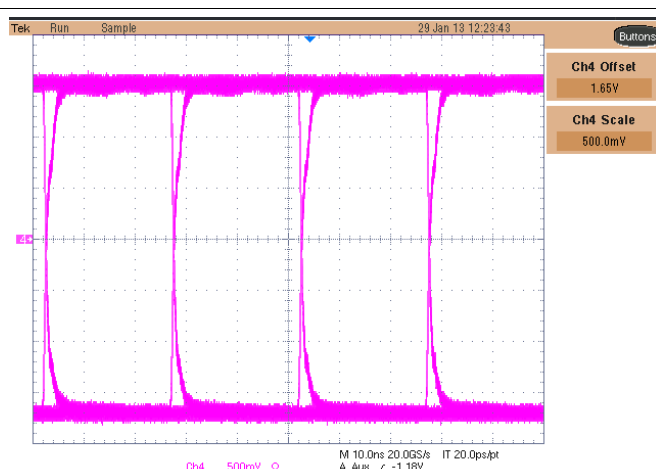


Figure 22. Typical Eye Diagram at 40 MBPS, PRBS 2¹⁶ - 1, 3.3-V Operation

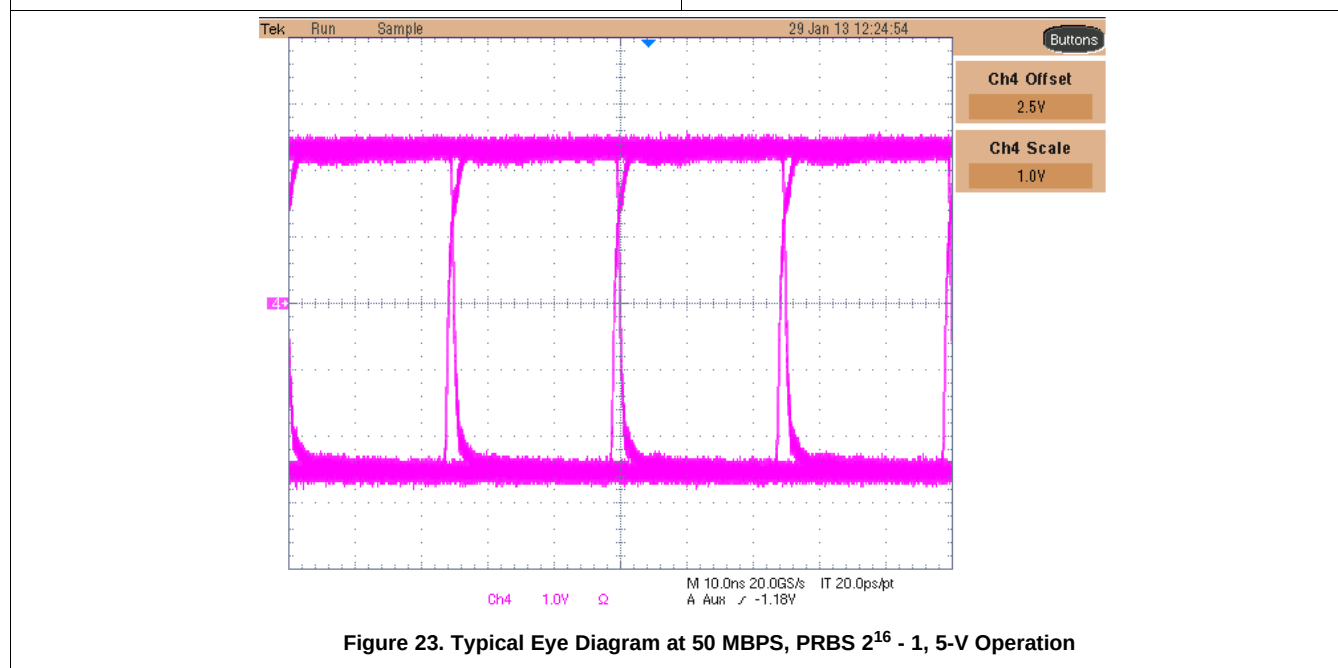


Figure 23. Typical Eye Diagram at 50 MBPS, PRBS 2¹⁶ - 1, 5-V Operation

Typical Applications (continued)

9.2.2 Isolated RS-485 Interface

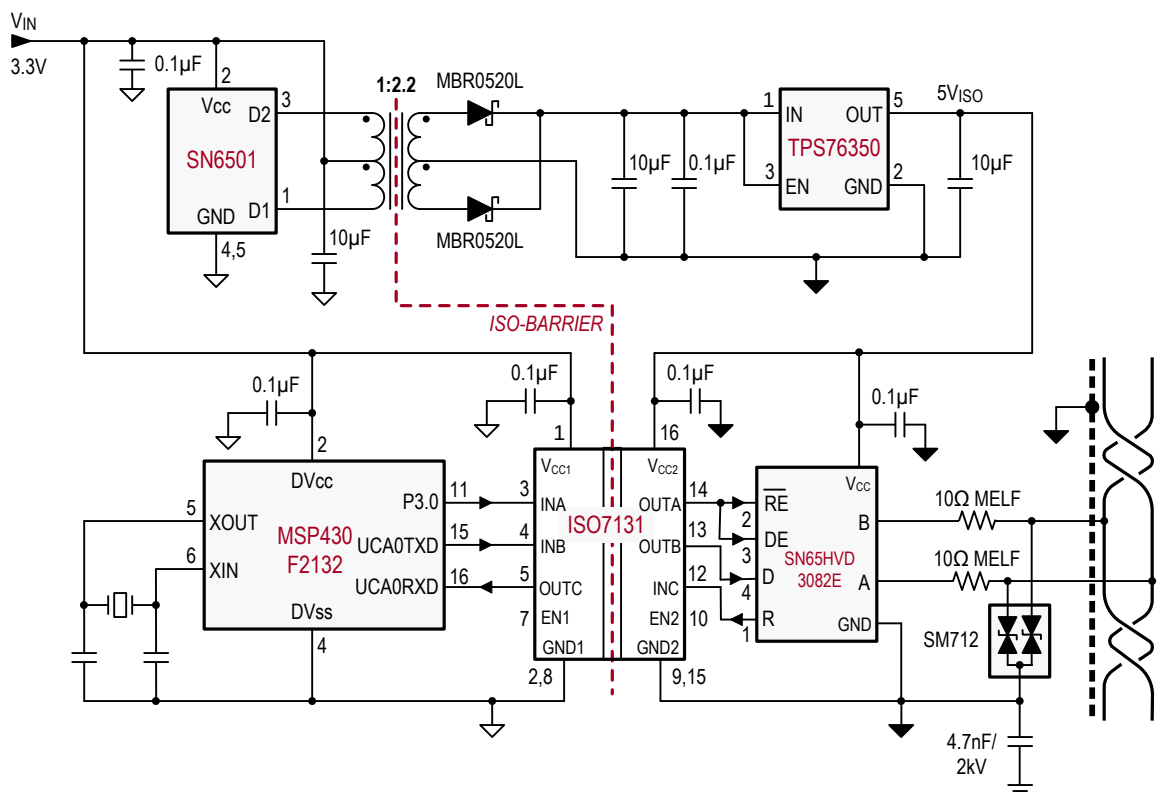


Figure 24. Isolated RS-485 Interface

9.2.2.1 Design Requirements

See previous *Design Requirements*.

9.2.2.2 Detailed Design Procedure

See previous *Detailed Design Procedure*.

9.2.2.3 Application Curves

See previous *Application Curves*.

10 Power Supply Recommendations

To ensure reliable operation at all data rates and supply voltages, a 0.1- μ F bypass capacitor is recommended at input and output supply pins (V_{CC1} and V_{CC2}). The capacitors should be placed as close to the supply pins as possible. If only a single primary-side power supply is available in an application, isolated power can be generated for the secondary-side with the help of a transformer driver such as TI's [SN6501](#). For such applications, detailed power supply design and transformer selection recommendations are available in [SN6501](#) data sheet ([SLLSEA0](#)).

11 Layout

11.1 Layout Guidelines

A minimum of four layers is required to accomplish a low EMI PCB design (see [Figure 25](#)). Layer stacking should be in the following order (top-to-bottom): high-speed signal layer, ground plane, power plane and low-frequency signal layer.

- Routing the high-speed traces on the top layer avoids the use of vias (and the introduction of their inductances) and allows for clean interconnects between the isolator and the transmitter and receiver circuits of the data link.
- Placing a solid ground plane next to the high-speed signal layer establishes controlled impedance for transmission line interconnects and provides an excellent low-inductance path for the return current flow.
- Placing the power plane next to the ground plane creates additional high-frequency bypass capacitance of approximately 100pF/in².
- Routing the slower speed control signals on the bottom layer allows for greater flexibility as these signal links usually have margin to tolerate discontinuities such as vias.

If an additional supply voltage plane or signal layer is needed, add a second power / ground plane system to the stack to keep it symmetrical. This makes the stack mechanically stable and prevents it from warping. Also the power and ground plane of each power system can be placed closer together, thus increasing the high-frequency bypass capacitance significantly.

For detailed layout recommendations, see Application Note [SLLA284](#), *Digital Isolator Design Guide*.

11.1.1 PCB Material

For digital circuit boards operating below 150 Mbps, (or rise and fall times higher than 1 ns), and trace lengths of up to 10 inches, use standard FR-4 epoxy-glass as PCB material. FR-4 (Flame Retardant 4) meets the requirements of Underwriters Laboratories UL94-V0, and is preferred over cheaper alternatives due to its lower dielectric losses at high frequencies, less moisture absorption, greater strength and stiffness, and its self-extinguishing flammability-characteristics.

11.2 Layout Example

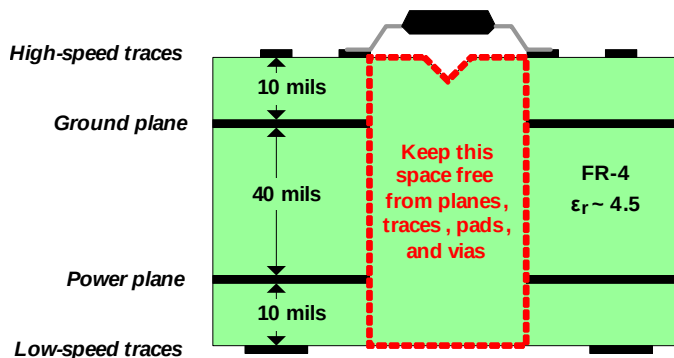


Figure 25. Recommended Layer Stack

12 Device and Documentation Support

12.1 Documentation Support

12.1.1 Related Documentation

- [SLLA284](#), *Digital Isolator Design Guide*
- [SLLSEA0](#), *Transformer Driver for Isolated Power Supplies*

12.2 Related Links

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to sample or buy.

Table 4. Related Links

PARTS	PRODUCT FOLDER	SAMPLE & BUY	TECHNICAL DOCUMENTS	TOOLS & SOFTWARE	SUPPORT & COMMUNITY
ISO7131CC	Click here	Click here	Click here	Click here	Click here
ISO7140CC	Click here	Click here	Click here	Click here	Click here
ISO7140FCC	Click here	Click here	Click here	Click here	Click here
ISO7141CC	Click here	Click here	Click here	Click here	Click here
ISO7141FCC	Click here	Click here	Click here	Click here	Click here

12.3 Trademarks

Modbus is a trademark of Gould Inc.
All other trademarks are the property of their respective owners.

12.4 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

12.5 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

[SLLA353](#) - *Isolation Glossary*.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
ISO7131CCDBQ	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7131CC
ISO7131CCDBQ.A	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7131CC
ISO7131CCDBQR	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7131CC
ISO7131CCDBQR.A	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7131CC
ISO7131CCDBQRG4	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7131CC
ISO7131CCDBQRG4.A	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7131CC
ISO7140CCDBQ	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140CC
ISO7140CCDBQ.A	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140CC
ISO7140CCDBQR	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140CC
ISO7140CCDBQR.A	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140CC
ISO7140CCDBQRG4	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140CC
ISO7140CCDBQRG4.A	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140CC
ISO7140FCCDBQ	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140FC
ISO7140FCCDBQ.A	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140FC
ISO7140FCCDBQG4	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140FC
ISO7140FCCDBQG4.A	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140FC
ISO7140FCCDBQR	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140FC
ISO7140FCCDBQR.A	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7140FC
ISO7141CCDBQ	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141CC
ISO7141CCDBQ.A	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141CC
ISO7141CCDBQR	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141CC
ISO7141CCDBQR.A	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141CC
ISO7141CCDBQRG4	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141CC
ISO7141CCDBQRG4.A	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141CC
ISO7141FCCDBQ	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141FC
ISO7141FCCDBQ.A	Active	Production	SSOP (DBQ) 16	75 TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141FC
ISO7141FCCDBQR	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141FC
ISO7141FCCDBQR.A	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141FC
ISO7141FCCDBQRG4	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141FC

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
ISO7141FCCDBQRG4.A	Active	Production	SSOP (DBQ) 16	2500 LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	7141FC

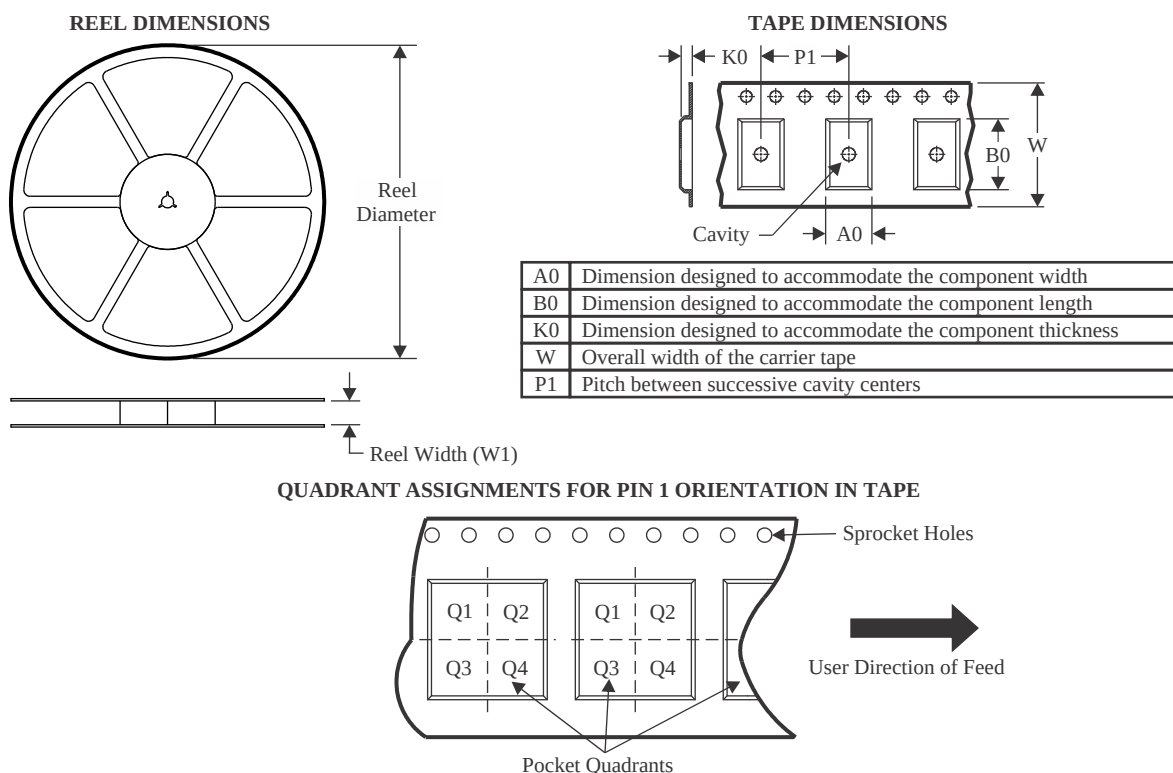
- ⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).
- ⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.
- ⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.
- ⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.
- ⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.
- ⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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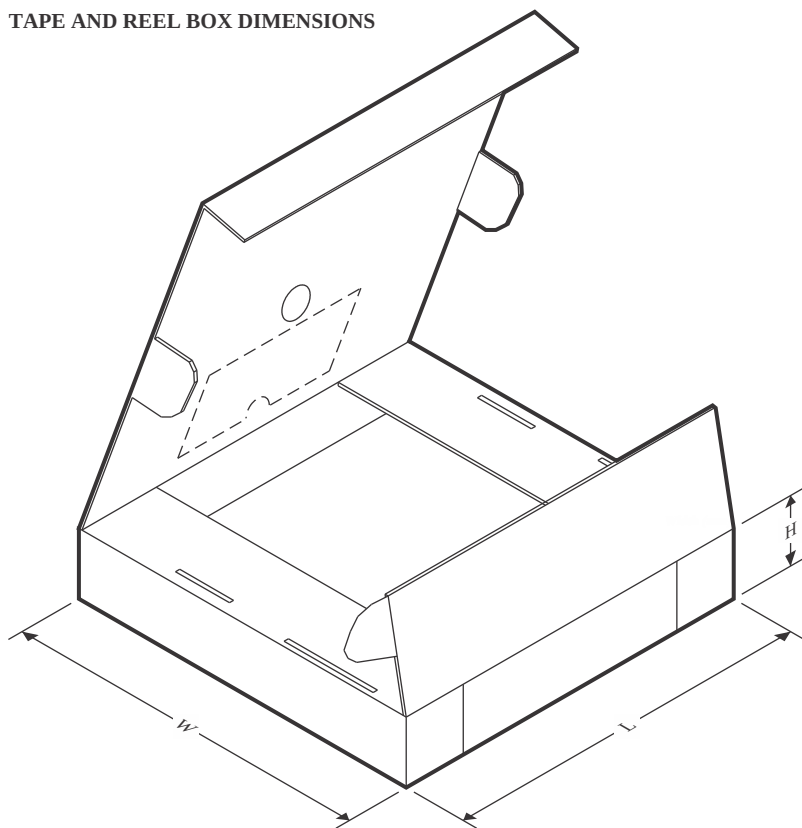
TAPE AND REEL INFORMATION



*All dimensions are nominal

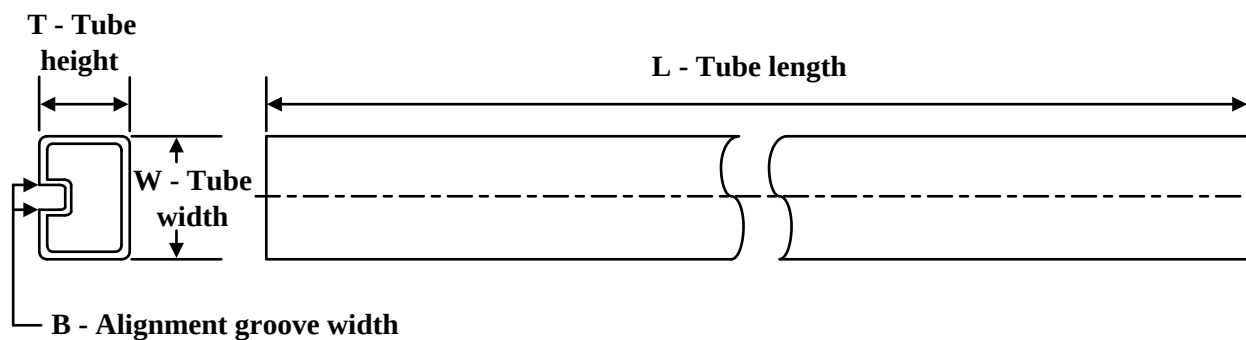
Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
ISO7131CCDBQR	SSOP	DBQ	16	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
ISO7131CCDBQRG4	SSOP	DBQ	16	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
ISO7140CCDBQR	SSOP	DBQ	16	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
ISO7140CCDBQRG4	SSOP	DBQ	16	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
ISO7140FCCDBQR	SSOP	DBQ	16	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
ISO7141CCDBQR	SSOP	DBQ	16	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
ISO7141CCDBQRG4	SSOP	DBQ	16	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
ISO7141FCCDBQR	SSOP	DBQ	16	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1
ISO7141FCCDBQRG4	SSOP	DBQ	16	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

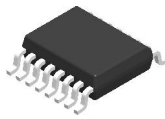
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
ISO7131CCDBQR	SSOP	DBQ	16	2500	350.0	350.0	43.0
ISO7131CCDBQRG4	SSOP	DBQ	16	2500	350.0	350.0	43.0
ISO7140CCDBQR	SSOP	DBQ	16	2500	350.0	350.0	43.0
ISO7140CCDBQRG4	SSOP	DBQ	16	2500	350.0	350.0	43.0
ISO7140FCCDBQR	SSOP	DBQ	16	2500	350.0	350.0	43.0
ISO7141CCDBQR	SSOP	DBQ	16	2500	350.0	350.0	43.0
ISO7141CCDBQRG4	SSOP	DBQ	16	2500	350.0	350.0	43.0
ISO7141FCCDBQR	SSOP	DBQ	16	2500	350.0	350.0	43.0
ISO7141FCCDBQRG4	SSOP	DBQ	16	2500	350.0	350.0	43.0

TUBE


*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
ISO7131CCDBQ	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7131CCDBQ	DBQ	SSOP	16	75	506.6	8	3940	4.32
ISO7131CCDBQ.A	DBQ	SSOP	16	75	506.6	8	3940	4.32
ISO7131CCDBQ.A	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7131CCDBQR	DBQ	SSOP	16	2500	506.6	8	3940	4.32
ISO7131CCDBQR.A	DBQ	SSOP	16	2500	506.6	8	3940	4.32
ISO7140CCDBQ	DBQ	SSOP	16	75	506.6	8	3940	4.32
ISO7140CCDBQ	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7140CCDBQ.A	DBQ	SSOP	16	75	506.6	8	3940	4.32
ISO7140CCDBQ.A	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7140CCDBQR	DBQ	SSOP	16	2500	506.6	8	3940	4.32
ISO7140CCDBQR.A	DBQ	SSOP	16	2500	506.6	8	3940	4.32
ISO7140FCCDBQ	DBQ	SSOP	16	75	506.6	8	3940	4.32
ISO7140FCCDBQ	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7140FCCDBQ.A	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7140FCCDBQ.A	DBQ	SSOP	16	75	506.6	8	3940	4.32
ISO7140FCCDBQG4	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7140FCCDBQG4.A	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7140FCCDBQR	DBQ	SSOP	16	2500	506.6	8	3940	4.32
ISO7140FCCDBQR.A	DBQ	SSOP	16	2500	506.6	8	3940	4.32
ISO7141CCDBQ	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7141CCDBQ	DBQ	SSOP	16	75	506.6	8	3940	4.32
ISO7141CCDBQ.A	DBQ	SSOP	16	75	506.6	8	3940	4.32
ISO7141CCDBQ.A	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7141CCDBQR	DBQ	SSOP	16	2500	506.6	8	3940	4.32
ISO7141CCDBQR.A	DBQ	SSOP	16	2500	506.6	8	3940	4.32
ISO7141FCCDBQ	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7141FCCDBQ	DBQ	SSOP	16	75	506.6	8	3940	4.32
ISO7141FCCDBQ.A	DBQ	SSOP	16	75	506.6	8	3940	4.32

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
ISO7141FCCDBQ.A	DBQ	SSOP	16	75	505.46	6.76	3810	4
ISO7141FCCDBQR	DBQ	SSOP	16	2500	506.6	8	3940	4.32
ISO7141FCCDBQR.A	DBQ	SSOP	16	2500	506.6	8	3940	4.32

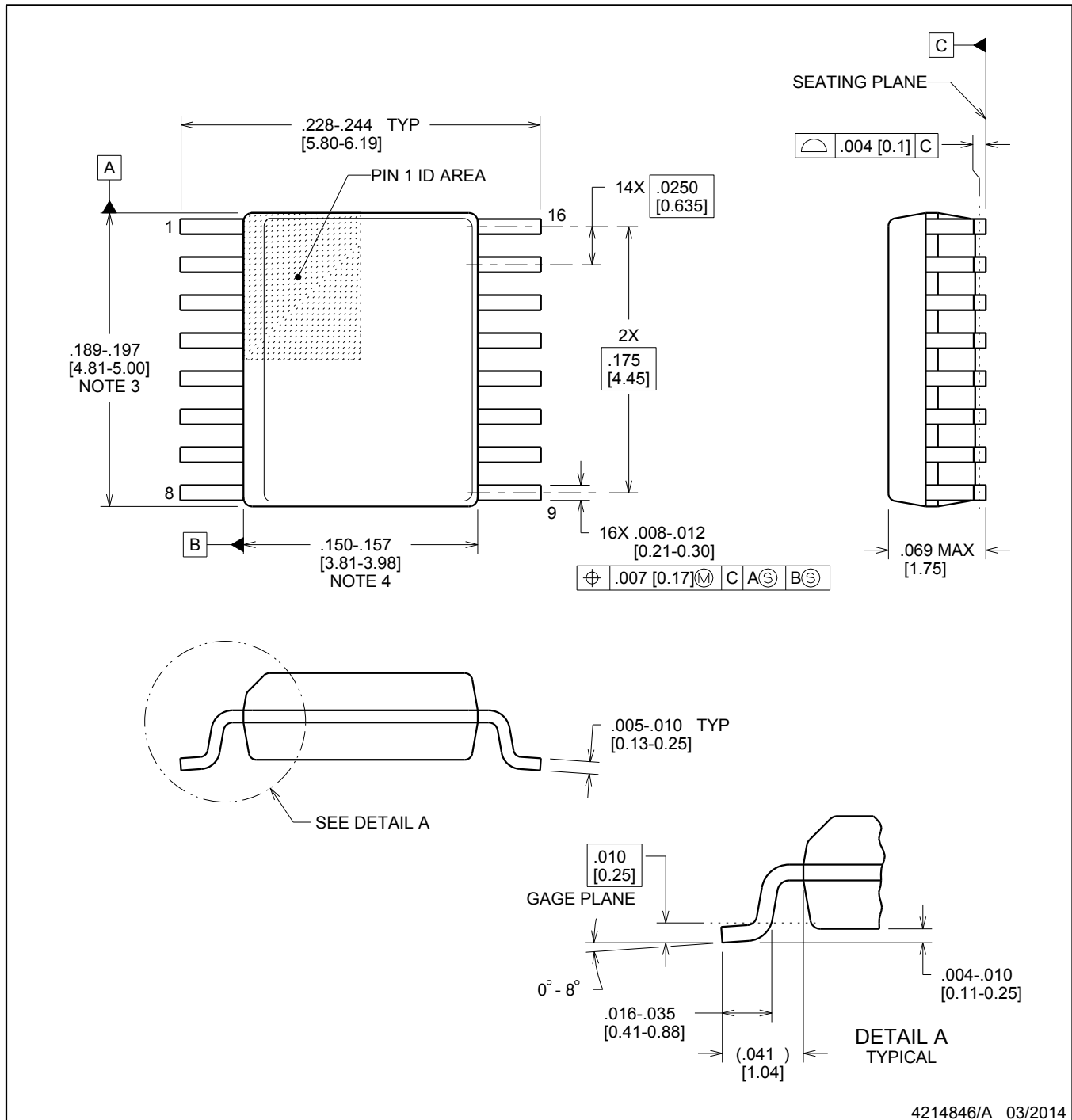


DBQ0016A

PACKAGE OUTLINE

SSOP - 1.75 mm max height

SHRINK SMALL-OUTLINE PACKAGE



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NOTES:

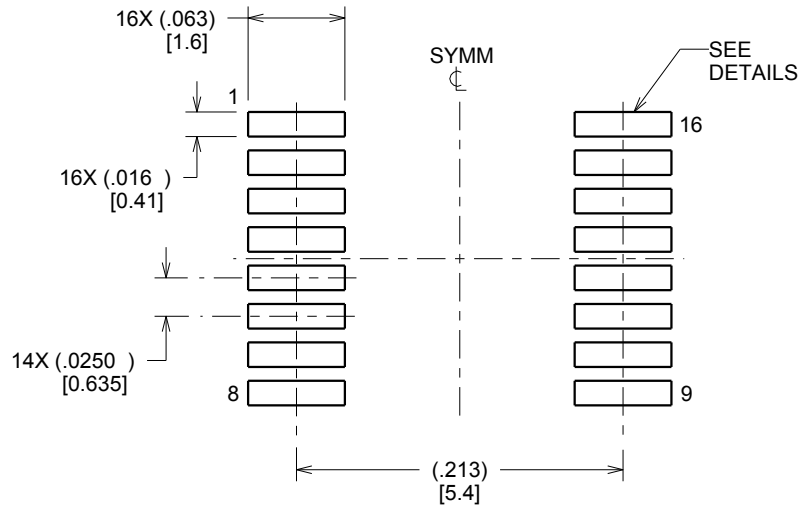
- Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
- This drawing is subject to change without notice.
- This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 inch, per side.
- This dimension does not include interlead flash.
- Reference JEDEC registration MO-137, variation AB.

EXAMPLE BOARD LAYOUT

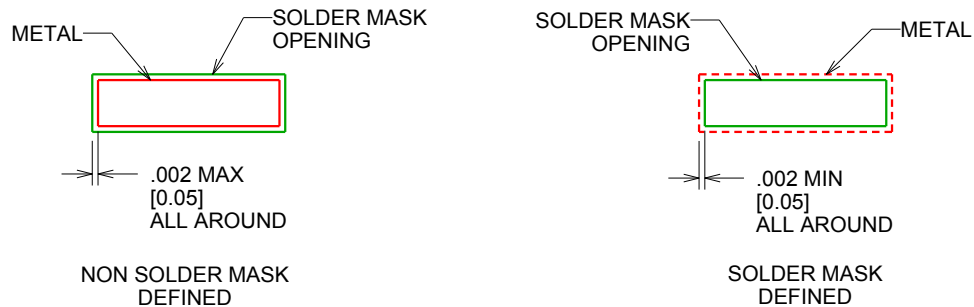
DBQ0016A

SSOP - 1.75 mm max height

SHRINK SMALL-OUTLINE PACKAGE



LAND PATTERN EXAMPLE
SCALE:8X



SOLDER MASK DETAILS

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NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

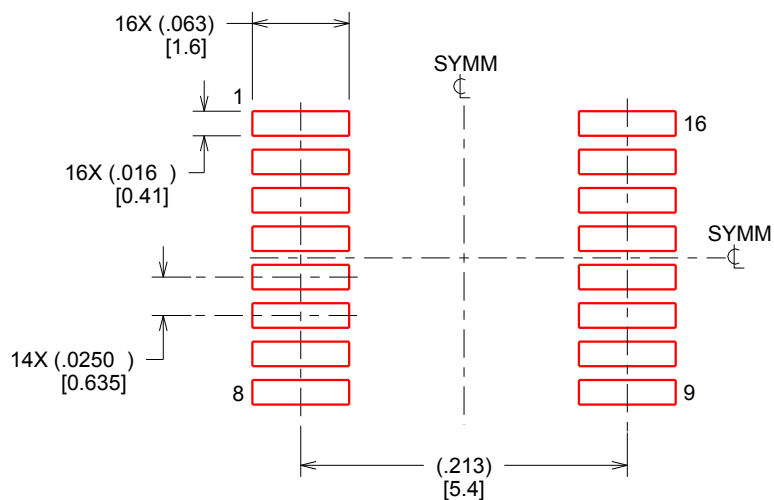
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

DBQ0016A

SSOP - 1.75 mm max height

SHRINK SMALL-OUTLINE PACKAGE



SOLDER PASTE EXAMPLE
BASED ON .005 INCH [0.127 MM] THICK STENCIL
SCALE:8X

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NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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